Pro Football Sports Science Research Update

August 2024

50 Years of Research on the Psychology of Sport Injury: A Consensus Statement. Sports Medicine, Ahead of Print: 1-16. 2024.

A systematic review of effects of ACL injury of soccer shoe outsoles, soccer playing field surfaces, and outsole-surface interface. Encyclopedia, 4: 888-899, 2024.

Acute myocardial infarction following "dry scooping" of a pre-workout supplement in a healthy young man of African origin: A case report. SAGE Open Medical Case Reports, Vol. 12: 1-6, 2024.

Age of first exposure does not relate to post-career health in former American-style football players. Sports Medicine, Ahead of Print: 1-10, 2024.

An examination of competitive rule evolution in the National Football League. Managing Sport and Leisure, Ahead of Print: 1-18, 2024.

Association between total genotype score and muscle injuries in top-level football players: A pilot study. Sports Medicine – Open, 10:22, 2024.

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Defining worst-case-scenario thresholds in soccer: Intensity vs volume. International Journal of Sports Physiology and Performance, Ahead of Print: 1-5, 2024.

Does it Matter? Isometric or isokinetic assessment of quadriceps strength symmetry 9 months after ACLR in collegiate athletes. Sports Health, Ahead of Print: 1-9, 2024.

Doping prevalence among U.S. elite athletes subject to drug testing under the World Anti-Doping Code. Sports Medicine – Open, 10:57, 2024.

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Effects of an 8-week pre-season targeted training on sprinting performance, agility and lower limb muscular asymmetries in elite soccer players. Biology of Sport, 41(4): 69-76, 2024.

Effects of multiple cold-water immersion during pre-season on recovery performance in under-20 male soccer players: A randomized controlled trial. Journal of Bodywork and Movement Therapies, Ahead of Print: 1-28, 2024.

Embracing the impact from instrumented mouthguards (iMGs): A survey of iMG mangers' perceptions of staff and player interest into the technology, data and barriers to use. European Journal of Sport Science, 24: 670-681, 2024.

Football movement profile-based creatine-kinase prediction performs similarly to global positioning system - Derived machine learning in national-team soccer players. International Journal of Sports Physiology and Performance, Ahead of Print: 1-8, 2024.

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50 Years of Research on the Psychology of Sport Injury: A Consensus Statement. Sports Medicine, Ahead of Print: 1-16. 2024.

Abstract - Factors influencing sport injury risk, rehabilitation outcomes, and return to sport processes have been the focus in various research disciplines (sports medicine, psychology and sociology). One discipline, with over 50 years of scholarship, is the psychology of sport injury. Despite the research in this field, there is no evidence-based consensus to inform professional practice. The aim of this original and timely consensus statement is to summarize psychological sport injury research and provide consensus recommendations for sport practitioners seeking to implement psychological principles into clinical practice. A total of seven experts with extensive experience outlined the consensus objectives and identified three psychology of sport injury sub-domains: risk, rehabilitation and return to sport. The researchers, grouped in pairs, prepared initial drafts of assigned sub-domains. The group met in Stockholm, and the three texts were merged into a draft and revised in an iterative process. Stress responses are the strongest psychological risk factor for acute injuries. Intra- and interpersonal factors, as well as sociocultural factors, are demonstrated psychosocial risk factors for overuse injuries. Stress management and mindfulness interventions to prevent injuries have been successfully implemented. The rehabilitation process may influence athlete's cognitive, emotional, and behavioral responses. Social support, mindfulness, acceptance-based practices, and cognitive-behavioral based intervention programs reduce negative reactions. Return to sport includes various stages and different trajectories. Returning athletes typically experience concerns regarding competence, autonomy, and relatedness. It is recommended that athletes focus on the physical, technical, and psychological demands of their sport as they progress to increasingly intense activities. Interdisciplinary collaboration (e.g., sports medicine and psychology) would be beneficial in enhancing clinical practice and improving athlete outcomes.

Key Points

- Regarding injury risk, form strong relationships with your athletes and colleagues, screen athletes frequently for psychosocial stress and perceived recovery and consider stakeholder education around organizational injury risk factors.
- Regarding rehabilitation, incorporate elements of mindfulness and acceptance-based practices and cognitive-behavioral based programs to improve coping and provide time and space to contribute as much social support as possible.

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- Regarding return to sport, utilize strategies that facilitate athletes' sense of competence, autonomy, relatedness and facilitate clear lines of communication between athletes, coaches and medical staff.

A systematic review of effects of ACL injury of soccer shoe outsoles, soccer playing field surfaces, and outsole-surface interface. Encyclopedia, 4: 888-899, 2024.

Abstract - Soccer is a sport with a high incidence of injuries. The most common injury occurs when the anterior cruciate ligament of the knee has undergone a sprain, strain, or partial or total rupture. Besides fitness factors (e.g., proprioception, balance and strength capabilities), a principal cause of ACL injuries is sport shoes and playing surfaces. Especially with the emergence of artificial surfaces (rubbers, turfs, concrete, asphalt, red turf), the ACL injuries dramatically increase. The cost of ACL injuries is high both in terms of career termination and in social and economic costs. Thus, it is necessary to understand the impact of sport shoes and playing surfaces on the ACL and how it can be mitigated. The present literature review followed the PRISMA methodology to identify the major biomechanical factors influencing the behavior of surfaces and shoes in relation to the ACL damage. Fifty-eight papers were identified. After reviewing the mechanism of injuries, we identified several distinct factors: type of outsole and mechanical arrangement of the outsoles; shear forces; characteristics of artificial turf surfaces; effect of weather on artificial surface aging and change in mechanical characteristics; outsole/ground interface; and the frictional, tractional and rotational forces in the pathogenesis of ACL injuries in soccer.

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Acute myocardial infarction following "dry scooping" of a pre-workout supplement in a healthy young man of African origin: A case report. SAGE Open Medical Case Reports, Vol. 12: 1-6, 2024.

Abstract - Dry scooping, a rather risky act of consuming undiluted pre-workout protein powder hoping for a more pronounced energy surge, is increasingly becoming a fitness trend worldwide. A previously healthy 25-year-old man of African origin presented with acute onset of crushing mid-sternal chest pain for 4h. His symptoms began about an hour following completion of his 2-h exercise regimen at the gym. He had an unremarkable past medical history except for an 8-month consumption of a pre-workout supplement as part of his gym routine. He claimed to have adhered to the manufacturer's recommended method of using the supplement, except for 3days prior to presentation when he started "dry scooping" following the viewing of a video advocating such practice from his fitness WhatsApp group. Cardiac evaluation revealed features in keeping with diagnosis of anterolateral ST-elevation myocardial infarction. Emergency coronary angiography revealed a thrombotic lesion completely occluding the proximal left anterior descending coronary artery with TIMI 0 flow distally. After successful balloon angioplasty, a drug-eluting stent was implanted successfully, restoring TIMI 3 flow to the proximal left anterior descending coronary artery. This case report aims to raise awareness among both clinicians and the general public regarding the pervasiveness of pre-workout supplements, improper methods of consumption, and the potential life-threatening repercussions of misuse.

Age of first exposure does not relate to post-career health in former American-style football players. Sports Medicine, Ahead of Print: 1-10, 2024.

Objective: Prior studies examining small samples of symptomatic former professional football players suggest that earlier age of first exposure (AFE) to American football is associated with adverse later life health outcomes. This study examined a larger, more representative sample of former professional American football players to assess associations between AFE before age 12 (AFE < 12) and clinical outcomes compared with those who started at age 12 or older (AFE 12 +).

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Methods: Former professional American football players who completed a questionnaire were dichotomized into AFE < 12 and AFE 12 +. AFE groups were compared on outcomes including symptoms of depression and anxiety, perceived cognitive difficulties, neurobehavioral dysregulation, and self-reported health conditions (e.g., headaches, sleep apnea, hypertension, chronic pain, memory loss, dementia/Alzheimer's disease, and others).

Results: Among 4189 former professional football players (aged 52 ± 14 years, 39% self-reported as Black), univariable associations with negligible effect sizes were seen with AFE < 12, depressive symptoms ($p = 0.03$; $\eta 2 = 0.001$), and anxiety-related symptoms ($p = 0.02$; $\eta 2 = 0.001$) only. Multivariable models adjusting for age, race, body mass index, playing position, number of professional seasons, and past concussion burden revealed no significant relationships between AFE < 12 and any outcome. Linear and non-linear models examining AFE as a continuous variable showed similar null results.

Conclusions: In a large cohort of former professional American-style football players, AFE was not independently associated with adverse later life outcomes. These findings are inconsistent with smaller studies of former professional football players. Studies examining AFE in professional football players may have limited utility and generalizability regarding policy implications for youth sports.

Key Points

- Prior studies suggested that starting to play football at younger ages (i.e., before 12 years old) may be associated with worse later-life outcomes (e.g., cognitive functioning, depression symptoms), but these studies had small sample sizes and all participants had self-reported cognitive, behavioral, and/or mood symptoms. Larger studies have not found this association.
- Questionnaire data provided by 4189 former American-style professional football players did not indicate that younger age of first exposure to football was significantly associated with any self-reported negative health outcomes when adjusting statistical models for other important variables.
- Future longitudinal cohort studies of youth football that further clarify the potentially harmful elements of youth football will assist parents and healthcare professionals in balancing the positive effects of football participation (e.g., cardiovascular conditioning, mentoring and community support, teamwork, self-esteem) and with the potential risks associated with play.

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An examination of competitive rule evolution in the National Football League. Managing Sport and Leisure, Ahead of Print: 1-18, 2024.

Purpose/rationale: Competitive rules, which have implications for player safety, fairness, and entertainment, change over time, yet institutional accounts offer only a high-level view of rule change. This study centers, and looks within, competitive rules to examine how they evolve in professional sport.

Design/methodology/approach: Using an inductive approach, we conducted a qualitative document analysis of NFL rule books published between 1980 and 2020. We developed and examined longitudinal trajectories of player conduct rules.

Findings: The analysis reveals rule components (i.e. actions, actors, times, spaces, sanctions) and patterns of change. Rules generally expanded in scope and length. We draw on communications from NFL leaders to connect the findings to institutional factors.

Practical implications: The findings offer a framework for rule managers as they iterate rules to make sport safer, fairer, and more entertaining.

Research contribution: This study sheds light on how competitive rules evolve at the level of component parts, providing a foundation for future institutional scholarship on the causes and consequences of rule change.

Originality/value: This research offers a microfoundations and longitudinal understanding of rule change in sport, with insights for academics and rule managers.

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Association between total genotype score and muscle injuries in top-level football players: A pilot study. Sports Medicine – Open, 10:22, 2024.

Background: Recently, genetic predisposition to injury has become a popular area of research and the association between a few single nucleotide polymorphisms (SNPs) and the susceptibility to develop musculoskeletal injuries has been shown. This pilot study aimed to investigate the combined effect of common gene polymorphisms previously associated with muscle injuries in Italian soccer players.

Results: A total of 64 Italian male top football players (age 23.1 ± 5.5 years; stature 180.2 ± 7.4 cm; weight 73.0 ± 7.9 kg) were genotyped for four gene polymorphisms [ACE I/D (rs4341), ACTN3 c.1729C > T (rs1815739), COL5A1 C > T (rs2722) and MCT1 c.1470A > T (rs1049434)]. Muscle injuries were gathered for 10 years (2009-2019). Buccal swabs were used to obtain genomic DNA, and the PCR method was used to genotype the samples. The combined influence of the four polymorphisms studied was calculated using a total genotype score (TGS: from 0 to 100 arbitrary units; a.u.). A genotype score (GS) of 2 was assigned to the "protective" genotype for injuries, a GS of 1 was assigned to the heterozygous genotype while a GS of 0 was assigned to the "worst" genotype. The distribution of genotype frequencies in the ACE I/D (rs4341), ACTN3 c.1729C > T (rs1815739) and MCT1 c.1470A > T (rs1049434) polymorphisms was different between non-injured and injured football players ($p = 0.001$; $p = 0.016$ and $p = 0.005$, respectively). The incidence of muscle injuries was significantly different among the ACE I/D (rs4341), ACTN3 c.1729C > T (rs1815739) and COL5A1 C > T (rs2722) genotype groups, showing a lower incidence of injuries in the "protective" genotype than "worse" genotype (ACE, $p < 0.001$; ACTN3, $p = 0.005$) or intermediate genotype (COL5A1, $p = 0.029$). The mean TGS in non-injured football players (63.7 ± 13.0 a.u.) was different from that of injured football players (42.5 ± 12.5 a.u., $p < 0.001$). There was a TGS cut-off point (56.2 a.u.) to discriminate non-injured from injured football players. Players with a TGS beyond this cut-off had an odds ratio of 3.5 (95%CI 1.8-6.8; $p < 0.001$) to suffer an injury when compared with players with lower TGS.

Conclusions: These preliminary data suggest that carrying a high number of "protective" gene variants could influence an individual's susceptibility to developing muscle injuries in football. Adapting the training load parameters to the athletes' genetic profile represents today the new frontier of the methodology of training.

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Biceps femoris long head stiffens after 2 weeks of training cessation in highly trained sprinters. European Journal of Applied Physiology, Ahead of Print: 1-7, 2024.

Purpose: Most athletes experience short-term training cessation because of illness, injury, post-season vacation, or other reasons. Passive muscle stiffness is a potential risk factor for a sprint-type hamstring strain injury, but limited information is available about the effect of short-term training cessation on passive muscle stiffness. The present study aimed to identify whether and how passive muscle stiffness of the biceps femoris long head (BFlh) would vary due to 2 weeks of training cessation in sprinters.

Methods: Passive BFlh shear-wave speed (a proxy for stiffness) was measured using ultrasound shear-wave elastography in 28 male sprinters, before and after 2 weeks of intervention. During the 2 weeks, the participants in the training-cessation group (n = 14) were allowed to maintain their normal daily activities but not to perform any physical training, including stretching and resistance exercises. The participants in the training continuation group (n = 14) performed the training (including maximum speed sprint, plyometric, and weight training) prescribed by their coaches 5 days per week.

Results: In the training-cessation group, passive BFlh shear-wave speed increased after the 2 weeks of training cessation (4.75 ± 0.77 to 5.00 ± 0.88 m/s, $P < 0.001$). In contrast, there was no significant difference before and after the 2 weeks of training continuation (4.90 ± 0.85 to 4.93 ± 0.85 m/s, $P = 0.521$).

Conclusions: The present findings indicate that muscles stiffen by training cessation in sprinting athletes.

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Defining worst-case-scenario thresholds in soccer: Intensity vs volume. International Journal of Sports Physiology and Performance, Ahead of Print: 1-5, 2024.

Purpose: This study aimed to enhance the understanding of soccer match peak demands by describing worst-case scenario (WCS) and time spent above 80% and 90% of the WCS for total distance (TD) and high-speed running (HSR). The investigation considered playing level (first team vs under-19 [U19] team) and playing position (center backs, fullbacks, midfielders, and forwards) to assess how WCS and the time spent above specific thresholds vary across different populations.

Methods: Data from 31 players in a professional Italian soccer club were collected during the 2022–23 season. Microtechnology devices tracked physical activity during matches. Players were categorized by position, and WCS was determined using rolling averages over a 1-minute period. Time spent above 80% and 90% of WCS for TD and HSR was calculated.

Results: The U19 team exhibited higher HSR WCS compared with the first team (~63 m·min−1 vs ~56 $m \cdot min^{-1}$). Midfielders recorded the highest TD WCS (~208 $m \cdot min^{-1}$), and forwards exhibited the highest HSR WCS (~70 m·min−1). The first team spent significantly more time above 80% (~6 min) and 90% (~1 min) of TD WCS. Midfielders spent significantly more time above the 80% (~7 min) of TD WCS, while forwards above the 80% (~2 min) of HSR WCS.

Conclusions: The study emphasizes that WCS used alone may not sufficiently capture real match intensity. Considering the time spent above specific thresholds provides additional insights (ie, between-levels differences and position). Practitioners should consider both WCS and time spent above thresholds for individualized training prescriptions, reflecting differences in playing roles.

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Does it Matter? Isometric or isokinetic assessment of quadriceps strength symmetry 9 months after ACLR in collegiate athletes. Sports Health, Ahead of Print: 1-9, 2024.

Background: Greater quadriceps strength symmetry is associated with better outcomes after anterior cruciate ligament reconstruction (ACLR). Isometric and isokinetic assessments of quadriceps strength inform therapeutic exercise prescription and return-to-sport decisions. It is unclear whether isometric and isokinetic measures provide similar information post-ACLR.

Hypothesis: Quadriceps strength symmetry is similar between isometric and isokinetic assessments. Isokinetic and isometric strength symmetries have similar associations to functional knee kinetics and self-reported knee function.

Study design: Cross-sectional study.

Level of evidence: Level 3.

Methods: NCAA Division I athletes (N = 35), 8.9 ± 2.5 months post-ACLR completed isometric and isokinetic quadriceps strength assessments, countermovement jumps (CMJs), and treadmill running. Self-reported knee function was assessed using the International Knee Documentation Committee Subjective Knee Form (IKDC). Agreement between isometric and isokinetic strength symmetry was assessed using Bland-Altman analysis, with associations to functional knee kinetics and IKDC assessed using Pearson correlations and linear regressions.

Results: Mean difference in quadriceps strength symmetry between isokinetic and isometric assessments was 1.0% (95% limits of agreement of -25.1% to 23.0%). Functional knee kinetics during running and CMJ were moderately to strongly associated with isometric strength symmetry ($r = 0.64$-0.80, $P < 0.01$) and moderately associated with isokinetic strength symmetry ($r = 0.41$-0.58, $P < 0.01$). IKDC scores were weakly to moderately associated with isometric ($r = 0.39$, $P = 0.02$) and isokinetic ($r = 0.49$, $P < 0.01$) strength symmetry.

Conclusion: Isokinetic and isometric assessments of quadriceps strength symmetry in collegiate athletes 9 months post-ACLR demonstrated strong agreement. Quadriceps strength symmetry is associated with functional knee kinetic symmetry post-ACLR.

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Clinical relevance: Considerable individual variation suggests mode of contraction should be consistent throughout postoperative assessment. Isometric strength symmetry may be a better indicator of functional knee kinetic symmetry, while isokinetic strength symmetry may be associated more closely with patient-reported outcomes.

Doping prevalence among U.S. elite athletes subject to drug testing under the World Anti-Doping Code. Sports Medicine – Open, 10:57, 2024.

Background: Determining the prevalence of doping within an elite athlete population is challenging due to the extreme sensitivity of the topic; however, understanding true doping prevalence is important when designing anti-doping programs and measuring their effectiveness. The objective of this study was to estimate the prevalence of doping among Olympic, Paralympic, World, and National-level competitive athletes in the United States subject to the World Anti-Doping Code. All athletes who were subject to the U.S. Anti-Doping Agency's Protocol for Olympic and Paralympic Movement Testing, a World Anti-Doping Code ("Code")-compliant anti-doping program, were invited to complete a web-delivered survey. Using a direct questioning approach, the survey items asked athletes whether they had used each specific category of banned substance / method on the World Anti-Doping Agency's Prohibited List. Multiple strategies to encourage honest reporting (e.g., protecting anonymity by collecting minimal demographic information; using an outside organization to administer the survey) and to detect inconsistent responses were used.

Results: Depending on the method of calculation, 6.5-9.2% of the 1,398 respondents reported using one or more prohibited substances or methods in the 12 months prior to survey administration. Specific doping prevalence rates for each individual substance / method categories ranged from 0.1% (for both diuretics / masking agents and stem cell / gene editing) to 4.2% for in-competition use of cannabinoids.

Conclusion: Determining the prevalence of doping within different athlete populations is critical so that sport governing bodies can evaluate their anti-doping efforts and better tailor their programming. By measuring doping prevalence of specific categories of substances and methods, rather than just the overall prevalence of doping, this study also highlights where sport governing bodies should focus their future educational and detection efforts.

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Effects of an 8-week pre-season targeted training on sprinting performance, agility and lower limb muscular asymmetries in elite soccer players. Biology of Sport, 41(4): 69-76, 2024.

Abstract - The purpose of this study was to determine the effects of an 8-week targeted training program on speed, agility, and muscle asymmetries in soccer players. 32 elite soccer players were recruited for the study. Their age, body mass, and body height were 25.8±7.3 years, 77.4±11.1 kg, and 177.5±9.8 cm, respectively. After the warm-up, participants performed two maximum 30 m sprints with a 5-minute rest interval between trials. After the linear sprint test, performed two repetitions of the COD randomized ZigZag test. The tests were performed at the beginning and at the end of the pre-season, which lasted for 8 weeks. EMG was measured bilaterally from the quadriceps, hamstrings, and gluteal muscles with shorts made of knitted fabric similar to elastic clothes. Athletes in the experimental group performed sport-specific targeted strength training based on movement patterns 4 times per week (Monday, Tuesday, Thursday, and Friday). The training included 6 bodyweight exercises (Bird Dog, Single-Leg Glute Bridge, Dead Bug, Side Plank, Reverse Lunge, and Clamshell), performed unilaterally in 5 sets of 10 repetitions of each exercise. The load progression included an additional set in each successive week of the experiment. The data analysis showed significant improvement in 5 m and 30 m sprints after applying the 8-week targeted training protocol. A statistically significant improvement in performance was also observed after the 8-week training period in case of COD, while the speed deficit also decreased significantly. The greatest improvements were observed during the COD test. As a result of repeated COD training over a period of 8 weeks, its technique was improved. Implementation of training methods, which target specific aspects of soccer in the pre-season training routines can improve key motor abilities for that sports discipline. A comprehensive training approach including speed, agility, and resistance training based on specific movement patterns should be applied by strength and conditioning practitioners in soccer teams to achieve peak physical performance and reduce injury risk due to the partial elimination of muscular asymmetries.

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Effects of multiple cold-water immersion during pre-season on recovery performance in under-20 male soccer players: A randomized controlled trial. Journal of Bodywork and Movement Therapies, Ahead of Print: 1-28, 2024.

Introduction: This study was designed to evaluate the effects of multiple cold-water immersions performed daily during the preseason period on biochemical, clinical, and neuromuscular aspects of muscle damage of soccer players.

Methods: This two-arm, prospectively registered, randomized controlled trial, blinded to statistician and assessors, was conducted at professional football club facilities. Twenty-three under-20 semi-professionals male soccer players were randomly allocated into cold water immersion group (bathtub with water and ice at 10 °C ± 1 °C for 10 min) or control group (rest for 10 min), every day, after training sessions during a preseason. Primary outcome was change in creatine kinase (CK) concentration, and secondary outcomes were changes in vertical jump performance, strength and perception of recovery at baseline (T0) and after protocol training (T1).

Results: Analysis of Covariance (ANCOVA) showed a statistically significant time-group interactions for CK concentration, with an average reduction of 280.39 U/L (CI95% = −519.14, −41.64; d = 0.55) in the cold-water immersion compared to the control group. No differences between groups were observed in any other measures.

Conclusion: Multiple cold-water immersions at 10 °C for 10 min decreases CK concentration but does not change any clinical and neuromuscular markers of muscle damage in soccer players during a 9-day preseason.

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Embracing the impact from instrumented mouthguards (iMGs): A survey of iMG mangers' perceptions of staff and player interest into the technology, data and barriers to use. European Journal of Sport Science, 24: 670-681, 2024.

Abstract - Instrumented mouthguards (iMGs) are a novel technology being used within rugby to quantify head acceleration events. Understanding practitioners' perceptions of the barriers and facilitators to their use is important to support implementation and adoption. This study assessed men's and women's rugby union and league iMG managers' perceptions of staff and player interest in the technology, data and barriers to use. Forty-six iMG managers (men's rugby union and league n = 20 and n = 9 and women's rugby union and league n = 7 and n = 10) completed an 18-question survey. Perceived interest in data varied across staff roles with medical staff being reported as having the most interest. The iMG devices were perceived as easy to use but uncomfortable. Several uses of data were identified, including medical applications, player monitoring and player welfare. The comfort, size and fit of the iMG were reported as the major barriers to player use. Time constraints and a lack of understanding of data were barriers to engagement with the data. Continued education on how iMG data can be used is required to increase player and staff buy-in, alongside improving comfort of the devices. Studies undertaken with iMGs investigating player performance and welfare outcomes will make data more useful and increase engagement.

Highlights

- From the perspective of instrumented mouthguard (iMG) managers (appointed practitioners responsible for the collection and analysis of iMG technology is easy to use and has important sport science, medical and player welfare applications, yet is rarely used for these purposes in applied practice.
- To overcome barriers to adoption, technology companies should work closely with athletes to optimize the comfort, size and fit of iMGs and look to address any technological short comings perceived by practitioners, such as robustness and data anomalies.
- Future research should focus on the use of iMGs for sport science, medical and welfare purposes to improve understanding of iMG data and guide practice.

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Football movement profile-based creatine-kinase prediction performs similarly to global positioning system - Derived machine learning in national-team soccer players. International Journal of Sports Physiology and Performance, Ahead of Print: 1-8, 2024.

Purpose: The relationship between external load and creatine-kinase (CK) response at the team/position or individual level using Global Positioning Systems (GPS) has been studied. This study aimed to compare GPS-derived and Football Movement Profile (FMP) –derived CK-prediction models for national-team soccer players. The second aim was to compare the performance of general and individualized CK prediction models.

Methods: Four hundred forty-four national-team soccer players (under 15 [U15] to senior) were monitored during training sessions and matches using GPS. CK was measured every morning from whole blood. The players had 19.3 (18.1) individual GPS-CK pairs, resulting in a total of 8570 data points. Machine learning models were built using (1) GPS-derived or (2) FMP-based parameters or (3) the combination of the 2 to predict the following days' CK value. The performance of general and individual-specific prediction models was compared. The performance of the models was described by R2 and the root-mean-square error (RMSE, in units per liter for CK values).

Results: The FMP model (R^2 = .60, RMSE = 144.6 U/L) performed similarly to the GPS-based model (R^2 = .62, RMSE = 141.2 U/L) and the combination of the 2 (R^2 = .62, RMSE = 140.3 U/L). The prediction power of the general model was better on average (R^2 = .57 vs R2 = .37) and for 73% of the players than the individualized model.

Conclusions: The results suggest that FMP-based CK-prediction models perform similarly to those based on GPS-derived metrics. General machine learning models' prediction power was higher than those of the individual-specific models. These findings can be used to monitor post-match recovery strategies and to optimize weekly training periodization.

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Pro Football Sports Science Research Update

Hamstrings hypertrophy is specific to the training exercise: Nordic hamstring versus lengthened state eccentric training. Medicine and Science in Sports and Exercise, Ahead of Print: 1-57, 2024.

Introduction: The hamstring muscles play a crucial role in sprint running, but are also highly susceptible to strain injuries, particularly within the biceps femoris long head (BFlh). This study compared the adaptations in muscle size and strength of the knee flexors, as well as BFlh muscle and aponeurosis size, after two eccentrically focused knee flexion training regimes: Nordic hamstring training (NHT) or lengthened state eccentric training (LSET, isoinertial weight-stack resistance in an accentuated hip-flexed position), to habitual activity (no training controls: CON).

Methods: 42 healthy young males completed 34 sessions of NHT or LSET over 12 weeks or served as CON (n = 14/group). MRI-measured muscle volume of seven individual knee flexors and BFlh aponeurosis area, and maximum knee flexion torque during eccentric, concentric and isometric contractions were assessed pre- and post-training.

Results: LSET induced greater increases in hamstrings (+18% vs +11%) and BFlh (+19% vs +5%) muscle volumes and BFlh aponeurosis area (+9% vs +3%) than NHT (all $P \leq 0.001$), with no changes after CON. There were distinctly different patterns of hypertrophy between the two training regimes, largely due to the functional role of the muscles; LSET was more effective for increasing the size of knee flexors that also extend the hip (2.2-fold vs NHT), whereas NHT increased the size of knee flexors that do not extend the hip (1.9-fold vs LSET; both $P \leq 0.001$). Changes in maximum eccentric torque differed only between LSET and CON (+17% vs +4%; $P = 0.009$), with NHT (+11%) in-between.

Conclusions: These results suggest that LSET is superior to NHT in inducing overall hamstrings and BFlh hypertrophy, potentially contributing to better sprint performance improvements and protection against hamstring strain injuries than NHT.

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Injury prevention in professional soccer players: Can re-warmup training be organized in the post-warmup and half-time during a soccer game in a short time? Journal of Sports Medicine and Physical Fitness, Ahead of Print: 1-8, 2024.

Introduction: Passive time, during the post-warmup and half-time, decreases athletes' performance and increases injury risk factors in the active phases of the soccer match.

Objective: This narrative review aims to research and synthesize existing evidence to identify brief re-warmup strategies that may find applicability in the post-warmup and half-time of a soccer match.

Evidence acquisition: The analysis was conducted on PubMed, Web of Science, PEDro, SPORTDiscus and Google Scholar. Due to the lack of evidence, no temporal time was established, preferring most up-to-date articles. The data were synthesized in relation to the objectives, following the Preferred Reporting Items for Systematic Review and Meta-Analysis (PRISMA) guidelines.

Evidence synthesis: Our research yielded a total of 753 studies, 463 related to re-warmup, 136 articles on re-warmup in soccer, and 154 articles on re-warmup during half-time. Of these, 26 met the inclusion objectives and were included in this research. Our findings confirm that a re-warmup can mitigate the decremental effect of static rest on performance. From the studies considered, we have identified a work that emphasizes how a 1-minute warmup of high-intensity exercise at speed corresponding to 90% of VO<inf>2max</inf> can prevent decreases related to passive time, in sprint performance and muscle strength, as well as improve muscle temperature. These results could apply to both post-warmup and half-time scenarios of a soccer match.

Conclusions: From this narrative review, it has been possible to highlight a one-minute high-intensity re-warmup that improved sprint performance, increased core temperature, and enhanced muscle activation, not leading to additional physiological or psychological fatigue.

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Pro Football Sports Science Research Update

Internal or external training load metrics: Which is best for tracking autonomic nervous system recovery and function in collegiate American football? Journal of Functional Morphology and Kinesiology, 9:5, 2024.

Abstract - Sport coaches increasingly rely on external load metrics for designing effective training programs. However, their accuracy in estimating internal load is inconsistent, and their ability to predict autonomic nervous system (ANS) deterioration is unknown. This study aimed to evaluate the relationships between internal and external training load metrics and ANS recovery and function in college football players. Football athletes were recruited from a D1 college in the southeastern US and prospectively followed for 27 weeks. Internal load was estimated via exercise cardiac load (ECL; average training heartrate (HR) × session duration) and measured with an armband monitor equipped with electrocardiographic capabilities (Warfighter MonitorTM (WFM), Tiger Tech Solutions, Miami, FL, USA). External load was estimated via the summation and rate of acceleration and decelerations as measured by a triaxial accelerometer using the WFM and an accelerometer-based (ACCEL) device (Catapult Player Load, Catapult Sports, Melbourne, Australia) worn on the mid-upper back. Baseline HR, HR variability (HRV) and HR recovery served as the indicators for ANS recovery and function, respectively. For HRV, two, time-domain metrics were measured: the standard deviation of the NN interval (SDNN) and root mean square of the standard deviation of the NN interval (rMSSD). Linear regression models evaluated the associations between ECL, ACCEL, and the indicators of ANS recovery and function acutely (24 h) and cumulatively (one- and two-week). Athletes (n = 71) were male and, on average, 21.3 ± 1.4 years of age. Acute ECL elicited stronger associations for 24 h baseline HR (R 2 0.19 vs. 0.03), HR recovery (R 2 0.38 vs. 0.07), SDNN (R 2 0.19 vs. 0.02) and rMSSD (R 2 0.19 vs. 0.02) compared to ACCEL. Similar results were found for one-week: 24 h baseline HR (R 2 0.48 vs. 0.05), HR recovery (R 2 0.55 vs. 0.05), SDNN (R 2 0.47 vs. 0.05) and rMSSD (R 2 0.47 vs. 0.05) and two-week cumulative exposures: 24 h baseline HR (R 2 0.52 vs. 0.003), HR recovery (R 2 0.57 vs. 0.05), SDNN (R 2 0.52 vs. 0.003) and rMSSD (R 2 0.52 vs. 0.002). Lastly, the ACCEL devices weakly correlated with ECL (rho = 0.47 and 0.43, $p < 0.005$). Our findings demonstrate that ACCEL poorly predicted ANS deterioration and underestimated internal training load. ACCEL devices may “miss” the finite window for preventing ANS deterioration by potentially misestimating training loads acutely and cumulatively.

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Pro Football Sports Science Research Update

Metabolic consequences of anabolic steroids, insulin, and growth hormone abuse in recreational bodybuilders: Implications for the World Anti-Doping Agency Passport. Sports Medicine – Open, 10:28, 2024.

Background: Hormonal doping in recreational sports is a public-health concern. The World Anti-Doping Agency (WADA) promoted the creation of the Athlete Biological Passport, aiming to monitor athlete's biological variables over time to facilitate indirect detection of doping. Detection tests for anabolic androgenic steroids (AAS) and growth hormone (GH) are available while insulin abuse cannot be revealed. We have determined in recreational bodybuilders the metabolic effects associated with different patterns of hormone abuse. All analyses were conducted using Statistical Package for Social Sciences (SPSS) 21.0 software (SPSS Chicago, IL).

Results: We have assessed plasma concentrations of selected metabolic markers and fatty acid content in erythrocyte membranes of 92 male bodybuilders and in 45 healthy controls. Hormonal abuse was identified by anonymous questionnaires. 43% (%) of recruited bodybuilders regularly abused hormones, i.e., anabolic androgenic steroids (95%) often associated with GH (30%) and/or insulin (38%). HDL-cholesterol was lower in insulin and/or GH abusers. Alanine (ALT) and aspartic (AST) transaminases were greater in hormone abusing bodybuilders than in non-doping bodybuilders and controls. Insulin doping was selectively associated with increased plasma ALT-to-AST ratio. In erythrocyte membranes, elongase activity (i.e., stearic-to-palmitic ratio) was lower in insulin and/or growth hormone doping, whereas increased Δ-9 desaturase activity (i.e., palmitoleic-to-palmitic ratio) was selectively associated with insulin doping.

Conclusions: In conclusion, our study demonstrates that insulin and GH abuse are characterized by multiple alterations of specific metabolic markers. Although further studies are needed to test whether longitudinal monitoring of selected metabolic marker such as muscle contraction time, HDL levels, ALT-AST ratio as well as the activities of selected enzymes (e.g. Δ-9 desaturase and elongase), could contribute to the detection of insulin and GH abuse in sport.

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Pro Football Sports Science Research Update

Muscle morphological changes and enhanced sprint running performance: A 1-year observational study of well-trained sprinters. European Journal of Sport Science, Ahead of Print: 1-12, 2024.

Abstract - Numerous cross-sectional studies have attempted to identify the muscle morphology required to achieve high sprint velocity. Our longitudinal study addressed an unanswered question of cross-sectional studies: whether hypertrophy of the individual trunk and thigh muscles induced by daily training (e.g., sprint, jump, and resistance training) is linked to an improvement in sprint performance within well-trained sprinters. Twenty-three collegiate male sprinters (100-m best time of 11.36 ± 0.44 s) completed their daily training for 1 year without our intervention. Before and after the observation period, the sprint velocities at 0–100 m, 0–10 m, and 50–60 m intervals were measured using timing gates. The volumes of 14 trunk and thigh muscles were measured using magnetic resonance imaging. Muscle volumes were normalized to the participants' body mass at each time point. Sprint velocities increased at the 0–100 m ($p < 0.001$), 0–10 m ($p = 0.019$), and 50–60 m ($p = 0.018$) intervals after the observation period. The relative volumes of the tensor fasciae latae, sartorius, biceps femoris long head, biceps femoris short head, semitendinosus, and iliacus were increased (all $p < 0.050$). Among the hypertrophied muscles, only the change in the relative volume of the semitendinosus was positively correlated with the change in sprint velocity at the 50–60 m interval ($p = 0.018$ and $\rho = 0.591$). These findings suggest that semitendinosus hypertrophy seems to be associated with sprint performance improvement within well-trained sprinters during the maximal velocity phase.

Highlights

- Over the past few decades, numerous cross-sectional studies have attempted to identify the muscle morphology required to achieve high sprint velocity, but the question remains as to whether hypertrophy of the individual trunk and thigh muscles is linked to an improvement in sprint performance within well-trained sprinters.
- After 1-year daily training in well-trained sprinters, the tensor fasciae latae, sartorius, biceps femoris long head, biceps femoris short head, semitendinosus, and iliacus were hypertrophied.
- Among the hypertrophied muscles, only semitendinosus hypertrophy was associated with sprint performance improvement within well-trained sprinters during the maximal velocity phase of the 100-m sprint.

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Muscle-to-bone ratio in NCAA Division 1 collegiate football players by position. Journal of Strength and Conditioning Research, Ahead of Print: 1-6, 2024.

Abstract - The purpose of this study was to compare the muscle-to-bone ratio (MBR) in National Collegiate Athletic Association Division I football players (collegiate football players [CFP]) to healthy, age-matched controls. In addition, we examined MBR in CFP by position. A total of 553 CFP and 261 controls had their total and regional lean mass (LM), fat mass (FM), and bone mineral content (BMC) determined by dual x-ray absorptiometry (DXA). College football players were categorized by positions defined as offensive linemen (OL), defensive linemen (DL), tight end, linebacker (LB), running back (RB), punter or kicker, quarterback (QB), defensive back (DB), and wide receiver (WR). There were significant differences between CFP and controls for total LM (80.1 ± 10.0 vs. 56.9 ± 7.8 kg), FM (22.2 ± 12.5 vs. 15.2 ± 7.1 kg), and BMC (4.3 ± 0.5 vs. 3.1 ± 0.5 kg). Although there were significant differences in body composition between CFP and controls, there was no significant differences in total MBR between CFP and controls (18.6 ± 1.4 vs. 18.8 ± 1.7). Regionally, CFP had significantly lower trunk MBR than controls (26.7 ± 2.7 vs. 28.7 ± 4.2), but no difference was seen in leg or arm MBR. Positional differences in CFP were noted as total MBR being significantly higher in DL (19.0 ± 1.4) than in DB (18.1 ± 1.3), WR (18.1 ± 1.3), and LB (18.2 ± 1.3). OL had a significantly higher total MBR (19.2 ± 1.3) than DB (18.1 ± 1.3), LB (18.2 ± 1.3), QB (18.1 ± 1.0), and WR (18.1 ± 1.3). In addition, RB had significantly higher total MBR (18.8 ± 1.3) than DB (18.1 ± 1.3) and WR (18.1 ± 1.3). This study may provide athletes and training staff with normative values when evaluating total and regional MBR with DXA.

Practical Applications - This study explored the use of DXA to determine total and regional MBR in CFP. Although an MBR range suggesting optimal muscular loading of the skeleton may exist, there is currently no normative data to identify such a range in CFP specifically. Thus, this article provides insight into the MBR across positions in CFP, as well as in comparison to a control population. Moreover, these data provide information regarding total and regional MBR for different positions. These data can be used by athletes and coaches as reference data when using DXA to measure total and regional MBR because these numbers will likely differ from others such as skinfold, girth, and circumference measures. Moreover, these data provide information regarding the total and regional MBR for each position. The similarity between mirrored positions observed in this study and our previous report on NFL athletes suggests that MBR may play a role in on-field positional requirements. The data presented in this report can be used to identify ideal ranges for each position in collegiate football players. However, it is

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important to clarify that the data presented here are averages and SD for each position. In practice, performance metrics must be considered because each individual will have their own optimal body type to maximize performance. In collegiate football players total and regional MBR provides an objective measure to monitor changes in muscle and bone masses. As athletes are encouraged to alter their body composition, the MBR can provide a matrix to monitor these alterations. This article provides normative data to compare players against the normal population and positional differences that may prove to be useful in evaluating players.

NFL Ghosts: A framework for evaluating defender positioning with conditional density estimation. arXiv:2406.17220: 1-27, 2024.

Abstract - Player attribution in American football remains an open problem due to the complex nature of twenty-two players interacting on the field, but the granularity of player tracking data provides ample opportunity for novel approaches. In this work, we introduce the first public framework to evaluate spatial and trajectory tracking data of players relative to a baseline distribution of "ghost" defenders. We demonstrate our framework in the context of modeling the nearest defender positioning at the moment of catch. In particular, we provide estimates of how much better or worse their observed positioning and trajectory compared to the expected play value of ghost defenders. Our framework leverages high-dimensional tracking data features through flexible random forests for conditional density estimation in two ways: (1) to model the distribution of receiver yards gained enabling the estimation of within-play expected value, and (2) to model the 2D spatial distribution of baseline ghost defenders. We present novel metrics for measuring player and team performance based on tracking data, and discuss challenges that remain in extending our framework to other aspects of American football.

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Pro Football Sports Science Research Update

Normalization of rapid force to peak force in an isometric hamstring assessment using force plates. Apunts Sports Medicine, 59: 100450, 2024.

Purpose: The aim of this investigation was to determine the reliability of normalizing rapid force (RF) production to peak force assessed during an isometric knee flexor assessment, and to present a novel method of classifying athletes' potential training needs within the 90–90 isometric hamstring assessment.

Procedures: Twenty elite female soccer players (age: 20.7 ± 4.7 years; height: 168.2 ± 5.5 cm; body mass: 62.8 ± 7.0 kg), with no recent (>6 months) history of hamstring strain injury, volunteered to participate in the study. Following a standardized warm-up, each participant performed three maximal isometric hamstring contractions, with their heel resting on a force plate, elevated on a box, to ensure that their hips and knees were at 90° Data was analyzed to determine peak for (PF), RF was established as force expressed at 100 ms (F100) and force expressed at 200 ms (F200), with force at each time-point subsequently normalized to a percentage of PF.

Findings: F100 and F200 normalized to PF demonstrated good absolute reliability (%CV = 6.12–7.62) and moderate relative reliability (ICC = 0.689–0.703). Concurrently observing PF and normalized F100 and F200 could provide clear training and monitoring goals.

Conclusions: Normalizing measures of RF production, including F100 and F200, to PF can be performed reliability. Therefore, could be tracked overtime to identify changes as an effect of training or for fatigue monitoring purposes. However, further research is required to determine how knee flexor force-time characteristics change in relation to focused training and how these characteristics change in response to fatiguing activities.

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Pro Football Sports Science Research Update

Potential moderators of the effects of blood flow restriction training on muscle strength and hypertrophy: A meta-analysis based on comparison with high-load resistance training. Sports Medicine – Open, 10:58, 2024.

Background: While it has been examined whether there are similar magnitudes of muscle strength and hypertrophy adaptations between low-load resistance training combined with blood-flow restriction training (BFR-RT) and high-load resistance training (HL-RT), some important potential moderators (e.g., age, sex, upper and lower limbs, frequency and duration etc.) have yet to be analyzed further. Furthermore, training status, specificity of muscle strength tests (dynamic versus isometric or isokinetic) and specificity of muscle mass assessments (locations of muscle hypertrophy assessments) seem to exhibit different effects on the results of the analysis. The role of these influencing factors, therefore, remains to be elucidated.

Objectives: The aim of this meta-analysis was to compare the effects of BFR- versus HL-RT on muscle adaptations, when considering the influence of population characteristics (training status, sex and age), protocol characteristics (upper or lower limbs, duration and frequency) and test specificity.

Methods: Studies were identified through database searches based on the following inclusion criteria: (1) pre- and post-training assessment of muscular strength; (2) pre- and post-training assessment of muscular hypertrophy; (3) comparison of BFR-RT vs. HL-RT; (4) score ≥ 4 on PEDro scale; (5) means and standard deviations (or standard errors) are reported or allow estimation from graphs. In cases where the fifth criterion was not met, the data were requested directly from the authors.

Results: The main finding of the present study was that training status was an important influencing factor in the effects of BFR-RT. The trained individuals may gain greater muscle strength and hypertrophy with BFR-RT as compared to HL-RT. However, the results showed that the untrained individuals experienced similar muscle mass gains and superior muscle strength gains in with HL-RT compared to BFR-RT.

Conclusion: Compared to HL-RT, training status is an important factor influencing the effects of the BFR-RT, in which trained can obtain greater muscle strength and hypertrophy gains in BFR-RT, while untrained individuals can obtain greater strength gains and similar hypertrophy in HL-RT.

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Pro Football Sports Science Research Update

Prevalence of sonographic Achilles tendon, patellar tendon, and plantar fascia abnormalities in Division 1 collegiate athletes from a variety of sports. Clinical Journal of Sport Medicine, Ahead of Print: 1-7, 2024.

Objective: This study aimed to determine the prevalence of ultrasound abnormalities in the Achilles tendon, patellar tendon, and plantar fascia among a large cohort of collegiate student-athletes.

Design: Observational cross-sectional study.

Setting: Three Division I institutions.

Participants: 243 student-athletes participated in this study. Exclusion criteria included those younger than 18 years or who underwent prior surgery/amputation of structures, including anterior cruciate ligament (ACL) surgeries with patellar tendon grafts.

Interventions: Ultrasound examination of the Achilles tendon, patellar tendon, and plantar fascia of each leg was performed. An experienced sonographer reviewed each tendon video in a blinded manner, with a separate experienced sonographer separately reviewing to establish inter-rater reliability.

Main outcome measures: The primary outcome measured was the presence of any sonographic abnormality including hypoechogenicity, thickening, or neovascularity.

Results: Ultrasound abnormalities were identified in 10.1%, 37.2%, and 3.9% of all Achilles tendons, patellar tendons, and plantar fasciae, respectively. Abnormalities were significantly associated with the presence of concurrent pain for all structures ($P < 0.01$). Specifically, athletes with sonographic abnormalities were approximately 4 times [relative risk (RR) = 4.25; 95% confidence interval (CI), 2.05-8.84], 6 times (RR = 5.69; 95% CI, 2.31-14.00), and 5 times (RR = 5.17; 95% CI, 1.76-15.25) more likely to self-report pain in the Achilles tendon, patellar tendon, and plantar fascia, respectively.

Conclusions: This multi-institutional study completed at 3 Division I institutions is the largest study of its kind to identify the prevalence of sonographic abnormalities in the Achilles tendon, patellar tendon, and plantar fascia among collegiate student-athletes of various sports.

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Pro Football Sports Science Research Update

Reliability of three landmine-punch-throw variations and their load-velocity relationships performed with the dominant and nondominant hands. International Journal of Sports Physiology and Performance, Ahead of Print: 1-8, 2024.

Purpose: This study assessed the reliability and load–velocity profiles of 3 different landmine-punch-throw variations (seated without trunk rotation, seated with trunk rotation, and standing whole body) with different loads (20, 22.5, and 25.0 kg), all with the dominant hand and nondominant hand.

Methods: In a quasi-randomized order, 14 boxers (24.1 [4.3] y, 72.6 [10.1] kg) performed 3 repetitions of each variation with their dominant hand and their nondominant hand, with maximal effort and 3 minutes of interset rest. Peak velocity was measured via the GymAware Power Tool (Kinetic Performance Technologies). The interclass correlation coefficients and their 95% CIs were used to determine the intrasession reliability of each variation × load × hand combination. Additionally, a 2 (hand) × 3 (variation) repeated-measures analysis of variance assessed the load–velocity profile slope, and a 3 (variation) × 2 (hand) × 3 (load) repeated-measures analysis of variance assessed the peak velocity of each variation.

Results: Most variations were highly reliable (intraclass correlation coefficient > .91), with the nondominant hand being as reliable or more reliable than the dominant hand. Very strong linear relationships were observed for the group average for each variation ($R^2 \geq .96$). However, there was no variation × hand interaction for the slope, and there was no main effect for variation or hand. Additionally, there was no interaction for the peak velocity, but there were main effects for variation, hand, and load ($P < .01$).

Conclusion: Each variation was reliable and can be used to create upper-body ballistic unilateral load–velocity profiles. However, as with other research on load–velocity profile, individual data allowed for more accurate profiling than group average data.

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Pro Football Sports Science Research Update

Single session intermittent heat exposure with more frequent and shorter cooling breaks facilitates greater training intensity and elicits physiological responses comparable to continuous heat exposure. International Journal of Sports Physiology and Performance, Ahead of Print: 1-11, 2024.

Purpose: To investigate the influence of shorter, more frequent rest breaks with per-cooling as an alternative heat-acclimation session on physiological, perceptual, and self-paced maximal cycling performance, compared with continuous heat exposure.

Methods: Thirteen participants completed 1 continuous and 3 intermittent-heat-exposure (IHE) maximal self-paced cycling protocols in a random order in heat (36 °C, 80% relative humidity): 1 × 60-minute exercise (CON), 3 × 20-minute exercise with 7.5-minute rest between sets (IHE-20), 4 × 15-minute exercise with 5-minute rest between sets (IHE-15), and 6 × 10-minute exercise with 3-minute rest between sets (IHE-10). Mixed-method per-cooling (crushed-ice ingestion and cooling vest) was applied during rest periods of all IHE protocols.

Results: Total distance completed was greater in IHE-10, IHE-15, and IHE-20 than in CON (+11%, +9%, and +8%, respectively), with no difference observed between IHE protocols. Total time spent above 38.5 °C core temperature was longer in CON compared with IHE-15 and IHE-20 (+62% and +78%, respectively) but similar to IHE-10 (+5%). Furthermore, a longer time above 38.5 °C core temperature occurred in IHE-10 versus IHE-15 and IHE-20 (+54% and +69%, respectively). Sweat loss did not differ between conditions.

Conclusion: IHE with per-cooling may be a viable alternative heat-acclimation protocol in situations where training quality takes precedence over thermal stimulus or when both factors hold equal priority.

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Pro Football Sports Science Research Update

The effect of cannabidiol on subjective responses to endurance exercise: A randomized controlled trial. Sports Medicine – Open, 10:61, 2024.

Background: Exercise is known to improve health. However, it can be unpleasant, often inducing negative feelings, or 'affect'. Cannabidiol (CBD), a non-intoxicating constituent of the cannabis plant, has been reported to enhance the subjective experience of exercise; specifically, in trained individuals performing fixed-intensity endurance activity. Here, we investigated the effects of CBD on subjective responses to exercise under more ecologically valid conditions; namely, in recreationally active individuals performing self-paced endurance activity.

Methods: A randomized, double-blind, placebo-controlled, crossover trial was conducted at Griffith University between July 17 and August 28, 2023. Griffith University students studying sports nutrition were invited to take part, with eligible volunteers ≥ 18 years of age and able to perform endurance exercise. Participants ingested placebo or 150 mg CBD in two soft-gel capsules 90 min before completing a self-paced 25-lap (10 km) run around an outdoor athletics track (400 m, synthetic). The primary outcomes were affective valence during exercise, assessed on completion of laps 6, 12, 18 and 24 using the 'Feelings Scale', and positive and negative affect, assessed at baseline, pre-run and post-run using the 'Positive and Negative Affect Schedule'. Exercise enjoyment, motivation and self-efficacy, the core features of the 'runner's high' (i.e., euphoria, pain, anxiety, sedation), perceived exertion and run time were also assessed.

Results: Fifty-two participants were randomized and 51 were included in the final sample (n = 22 female; 22 [21-25] years). Exercise induced negative affect (i.e., at the time of undertaking) and increased pain. CBD did not counteract either response. In fact, CBD had no significant effects on any of the outcomes measured. In contrast, exercise, once completed, increased positive affect, and decreased negative affect and anxiety.

Conclusions: CBD (150 mg, oral) does not appear to enhance the subjective experience of self-paced endurance exercise in recreationally active individuals. Nor, however, does it appear to compromise it. These findings suggest that CBD use is safe under exercise conditions and unlikely to impede physical activity participation. Our study also reaffirms the powerful mood-enhancing effects of exercise.

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The effects of mobile phone dependence on athletic performance and its mechanisms. Frontiers in Psychology, 15:1391258, 2024.

Abstract - Mobile phone dependence (also known as internet dependence, MPD), defined as a problematic behavior characterized by excessive use or intermittent craving to use a mobile phone, results in various social, behavioral, and affective problems in daily life. In sports, MPD is directly related to the physical and mental health and sports performance of athletes. The individual and environmental factors, neurobiological mechanisms and theoretical models of MPD affecting athletic performance were analyzed by reviewing previous studies, aiming to construct effective training and development protocols to prevent and control the occurrence of MPD in athletes. At present, athletic performance can be affected by MPD through individual factors and environmental factors. The neurobiological mechanisms between the two are based on the brain reward system and microwave radiation from mobile phones, with athletic performance being restricted by alterations in the corresponding brain regions. Relevant theoretical models mainly include the social cognitive model of self-regulation and the integrative model of self-control, which explain the interrelationship between MPD and athletic performance from the perspectives of athletes' self-regulation and self-control, respectively. As an emerging phenomenon, the influence pathways and mechanisms by which MPD affects athletic performance need to be further investigated. A longitudinal perspective should be adopted to trace the dynamic impact relationship between the two, and developing relevant theoretical frameworks from an interdisciplinary research perspective should be valuable for providing theoretical support for coaches and sports administrators to formulate scientific training protocols and thus improve the mental health of athletes.

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Pro Football Sports Science Research Update

The effects of velocity loss thresholds during resistance training on lower-limb performance enhancement: A systematic review with meta-analysis. International Journal of Sports Science & Coaching, Ahead of Print: 1-15, 2024.

Abstract - This study assessed the effects of squat training with different velocity loss (VL) thresholds on squat strength, heavy-load squat velocity (HLSV), light-load squat velocity (LLSV), countermovement jump (CMJ) height, and sprint time (ST). A systematic search of electronic databases was conducted. Meta-analysis was used to examine effects of trainings with different VL thresholds and used meta-regression to examine the interaction effects (correlation) between training effects and VL in different performance-based outcomes. Our systematic search yielded 1017 articles, 13 of which were included in the present study. Meta-analysis revealed that resistance training with both low (0%–20%) and high (30%–45%) VL thresh olds improved squat strength, HLSV, LLSV, and CMJ height, but only low VL improved ST ($p < 0.05$). Training gains for HLSV, LLSV, CMJ ($p=0.088$), and ST with low VL were superior to those with high VL ($p < 0.05$). Meta-regression revealed that CMJ ($p=0.076$) and ST ($p = 0.010$) demonstrated improvements as VL decreased in the range of 0% to 45%. In conclusion, squat resistance training with low (≤20%) VL provides relatively more effective training stimuli, improving performance in power and speed tasks. Optimal VL ranges may be in the range 10% to 20%, ≤20%, and ≤10% for the HLSV and LLSV, CMJ, and ST, respectively.

The relationship between multiple concussions and multidimensional sleep quality in collegiate-aged, active athletes. Sleep Health: Journal of the National Sleep Foundation, Ahead of Print: 1-8, 2024.

Objectives: Determine the association of cumulative concussion and repetitive head impacts with self-reported sleep quality in healthy collegiate-aged athletes.

Methods: Collegiate-aged athletes (N = 212; mean age 21.00, 62.7% male) completed semistructured interviews for sport and concussion history and the Pittsburgh Sleep Quality Index (PSQI). Number of concussions was retrospectively determined based on the 1993 American Congress of Rehabilitation Medicine (ACRM) criteria; repetitive head impact was measured based on the cumulative years of

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contact sport exposure. Associations of number of concussions and repetitive head impact exposure with global PSQI score, overall poor (PSQI >5) vs. good sleep, and binarized subscale scores were tested. Secondary analyses were conducted using alternative concussion criteria and metrics of repetitive head impact.

Results: The number of prior concussions was associated with higher PSQI global scores (B(SE)= 0.50(0.13), $p < .001$). Participants with more concussions were more likely to be poor sleepers (OR=1.52, $p < .001$), report poorer sleep quality (OR=1.29, $p = .037$), longer sleep latency (OR=1.34, $p = .005$), more sleep disturbances (OR=1.56, $p = .001$), increased use of sleep medications or sleep aids (OR=1.35, $p = .008$), and more sleep-related daily dysfunction (OR=1.38, $p = .002$). Similar results were observed for alternative definitions of concussion. No metric of repetitive head impact was associated with any sleep quality metric.

Conclusions: More prior concussions, but not repetitive head impact exposure, are associated with worse self-reported sleep, with subscale analyses showing concussion history associated with multiple aspects of subjective sleep quality rather than sleep quantity. Sleep represents an important factor to consider for future research aimed at characterizing and ultimately preventing adverse long-term health outcomes associated with concussion history.

The role of virtual reality in the management of football injuries. Medicina, 60:1000, 2024.

Abstract - Injuries represent a serious concern for football players, with a significant loss in terms of sport participation and long periods of rehabilitation. According to the 2019/20 UEFA Élite Club Injury Report, the average incidence of injuries during training is 2.8 per 1000 h of training, with an average absence from training of 20 days. In addition, injured athletes are 4 to 7 times more likely to relapse than uninjured athletes. High workloads and reduced recovery periods represent two of the most important modifiable risk factors. In this context, prevention and an adequate rehabilitation protocol are vital in managing injuries, reducing their incidence, and improving the return to competition. In recent years, technological development has provided new tools in rehabilitation, and Virtual reality (VR) has shown interesting results in treating neurologic and orthopedic pathologies. Virtual Reality (VR)

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technology finds application in the sports industry as a tool to examine athletes' technical movements. The primary objective is to detect the biomechanical risk factors associated with anterior cruciate ligament injury. Additionally, VR can be used to train athletes in field-specific techniques and create safe and controlled therapeutic environments for post-injury recovery. Moreover, VR offers a customizable approach to treatment based on individual player data. It can be employed for both prevention and rehabilitation, tailoring the rehabilitation and training protocols according to the athletes' specific needs.

The use of whole-body cryotherapy (WBC) for musculoskeletal ailments is growing. WBC, involving brief exposure to extremely low temperatures, is increasingly used for its analgesic, anti-infammatory, and antioxidant efects. The paper examines the physiological impacts of WBC on cardiovascular, musculoskeletal, hematologic, hormonal, and metabolic systems. Specifc orthopaedic applications discussed include its role in the management of fractures, osteoarthritis, osteone crosis of the femoral head, osteomyelitis, adhesive capsulitis, tendinopathies, rheumatic pathologies, chronic pain syndromes, and fbromyalgia. The study also highlights the benefts and drawbacks of WBC, including its potential to improve athletic performance, recovery, mood, and well-being, while noting risks like frostbite and impaired muscle strength. Clinical evi dence from various studies is evaluated, revealing a spectrum of outcomes. For instance, WBC shows promise in enhancing bone health in athletes and providing relief in osteoarthritis and fbromyalgia patients. However, evidence for its efcacy in conditions like chronic pain syndromes and osteomyelitis is less robust. The paper underscores the need for further research to establish standardised protocols and understand long-term efects. While WBC ofers signifcant benefts in orthopaedics, understanding its

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limitations and potential risks is crucial for its safe and efective clinical application. **Abstract -** The use of whole-body cryotherapy (WBC) for musculoskeletal ailments is growing. WBC, involving brief exposure to extremely low temperatures, is increasingly used for its analgesic, anti-inflammatory, and antioxidant effects. The paper examines the physiological impacts of WBC on cardiovascular, musculoskeletal, hematologic, hormonal, and metabolic systems. Specific orthopaedic applications discussed include its role in the management of fractures, osteoarthritis, osteonecrosis of the femoral head, osteomyelitis, adhesive capsulitis, tendinopathies, rheumatic pathologies, chronic pain syndromes, and fibromyalgia. The study also highlights the benefits and drawbacks of WBC, including its potential to improve athletic performance, recovery, mood, and well-being, while noting risks like frostbite and impaired muscle strength. Clinical evidence from various studies is evaluated, revealing a spectrum of outcomes. For instance, WBC shows promise in enhancing bone health in athletes and providing relief in osteoarthritis and fibromyalgia patients. However, evidence for its efficacy in conditions like chronic pain syndromes and osteomyelitis is less robust. The paper underscores the need for further research to establish standardized protocols and understand long-term effects. While WBC offers significant benefits in orthopaedics, understanding its limitations and potential risks is crucial for its safe and effective clinical application.

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